

Can we talk to the brain through the skin?

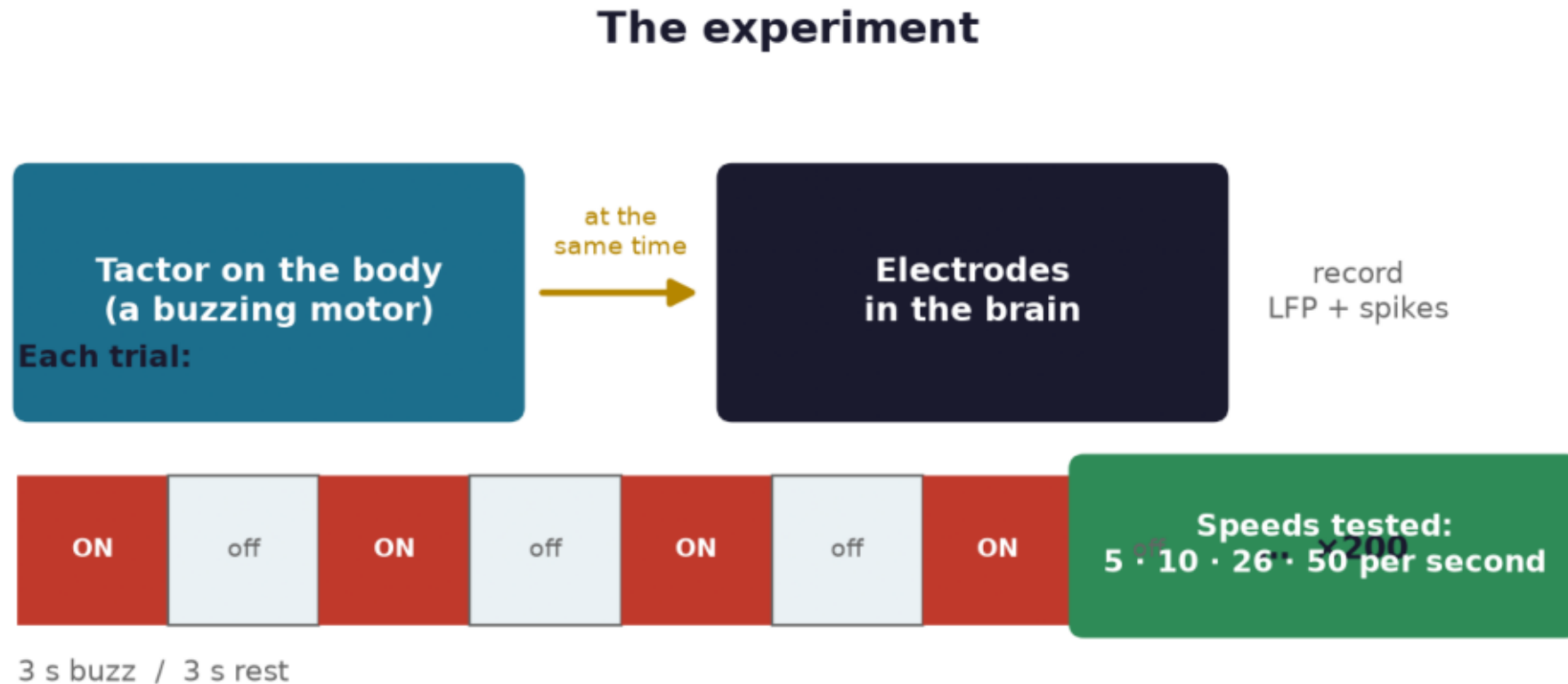
What happens in the brain when we buzz the body at different rhythms

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The dream: wearables that gently 'nudge' the brain through touch

To build that, we first have to answer a basic question: does a buzz on the body actually reach and change the brain?

The experiment, in one picture



Buzz the body in 3-second bursts at different speeds; listen to the brain at the same time; repeat 200×.

Three questions

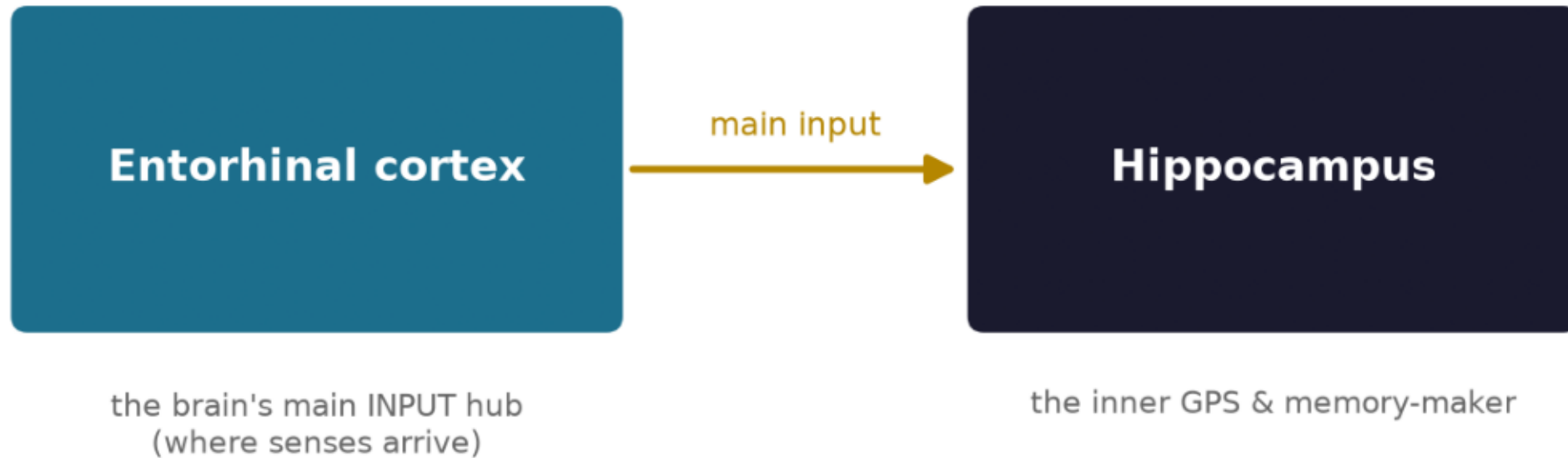
(1) Does the brain notice? (2) Does the buzz SPEED matter? (3) Does the brain march in step with the buzz?

Part 1 — How do you 'listen' to a brain?

Two ideas you need, both with analogies

We listened in two neighboring brain regions

Two connected listening posts — we recorded both

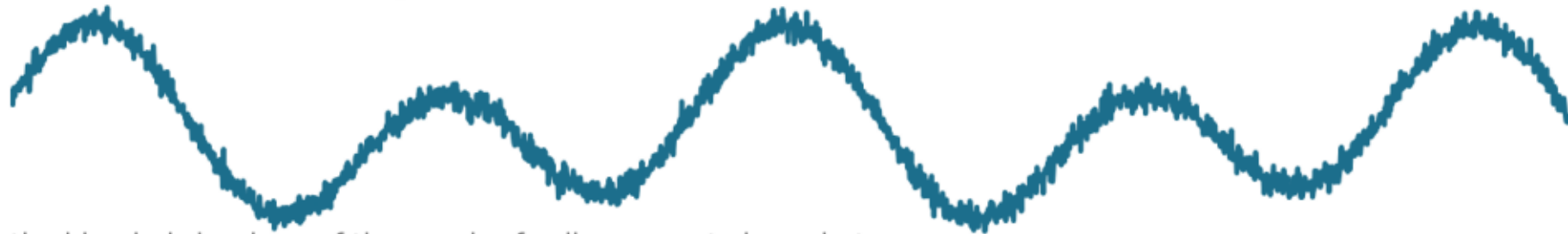


Hippocampus = the brain's inner GPS & memory. Entorhinal cortex = its main input hub. Two listening posts.

THE key idea: a brain recording has two very different signals

One electrode hears TWO things

CROWD ROAR (field potential, LFP)



the blended slow hum of thousands of cells — easy to hear, but vague and easily fooled by background noise.

INDIVIDUAL VOICES (single neurons firing = spikes)



each is a real, unambiguous event the machine's hum can't fake. ← our gold standard.

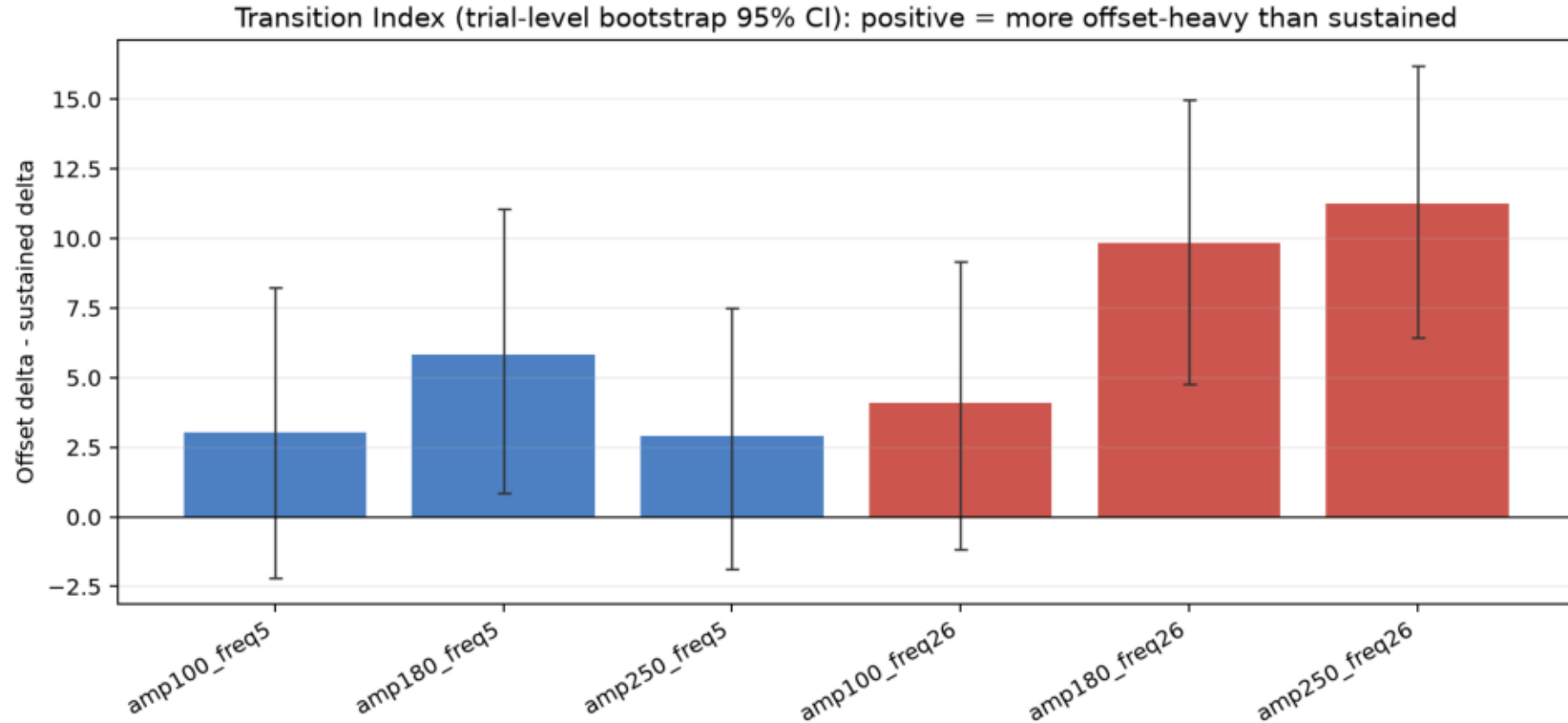
Stadium analogy → the CROWD MURMUR (the field potential) vs. INDIVIDUAL VOICES (single neurons firing).

Why single neurons are the gold standard

A real neuron firing is unambiguous — much harder for noise or equipment to fake than the crowd hum.

Part 2 — What the brain actually did

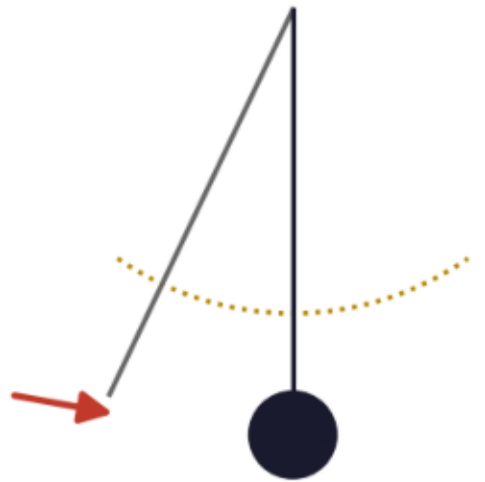
Yes — the brain notices the buzz. But it reacts to the **START** and **STOP**



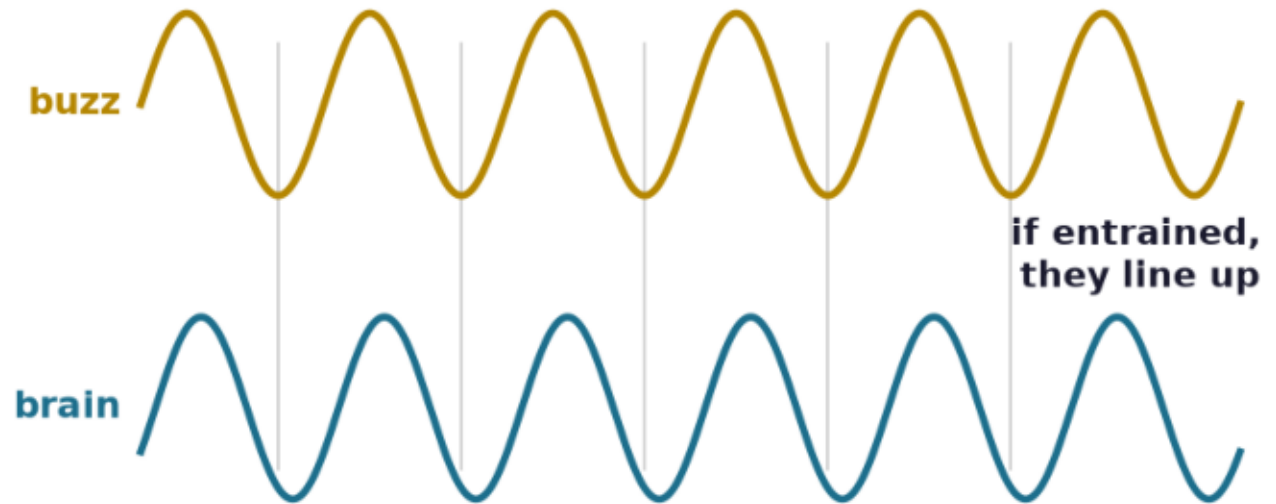
Like a doorbell: the brain responds to the **EVENT** of the buzz turning on/off — not by humming along with it.

'Marching in step' has a name: entrainment

Entrainment = the brain 'marching in step' with the buzz



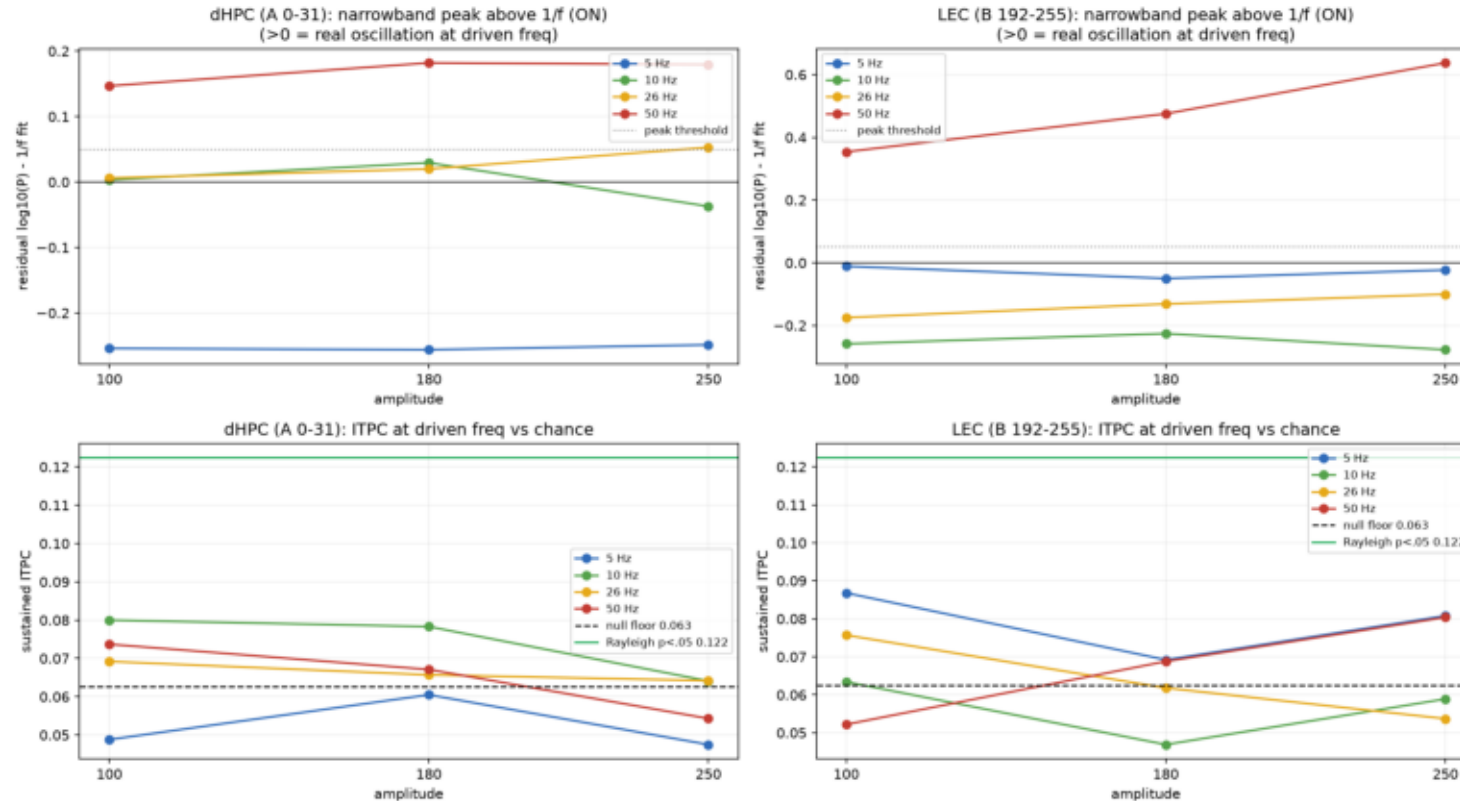
push in rhythm →
the swing goes higher



Pushing a swing in rhythm → if the brain entrains, its own waves line up with the buzz's beat.

The hippocampus does NOT hum along — at any buzz speed

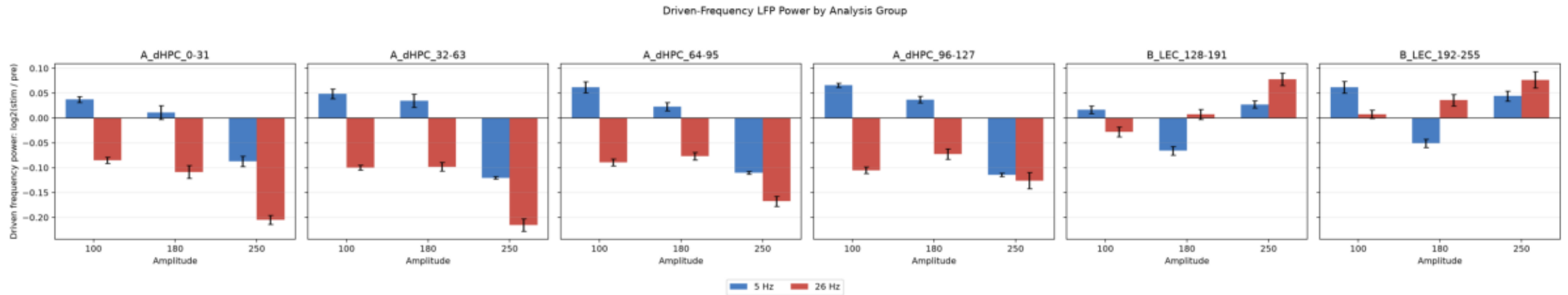
Dec 4 entrainment test: is the driven-frequency LFP a real oscillation (peak above 1/f) and/or phase-locked (ITPC > floor)?



No matter the buzz speed (5-50/sec), we see no matching brain rhythm above the normal background.

FINDING 3

Plot twist: the OTHER region shows a 50-per-second signal



In the entorhinal region a 50 Hz signal appears in the crowd murmur — and it grows with buzz strength. Exciting!

The trap: the machine has its OWN electrical hum

The trap: the machine has its OWN 50 Hz hum

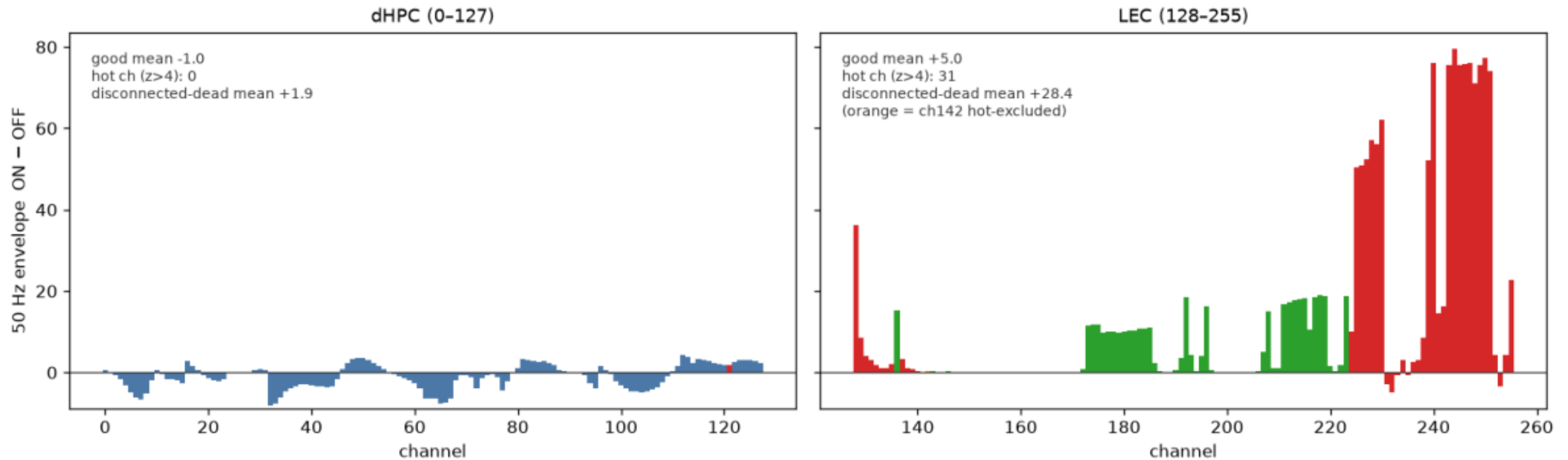


The mic records BOTH → a clean '50 Hz' could be the SPEAKER, not the brain.

A microphone can pick up the speaker's buzz, not just the crowd. Electrical equipment leaks its own signal.

That 50 Hz is mostly the machine — the dead-electrode test proves it

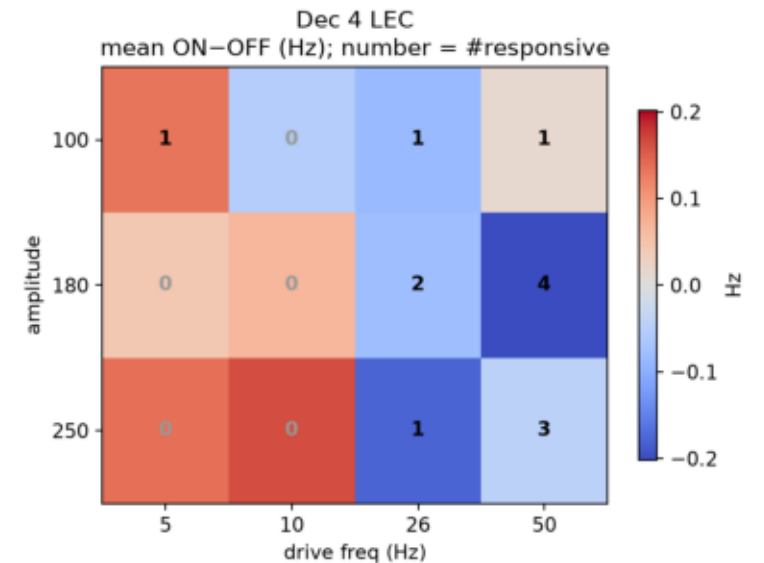
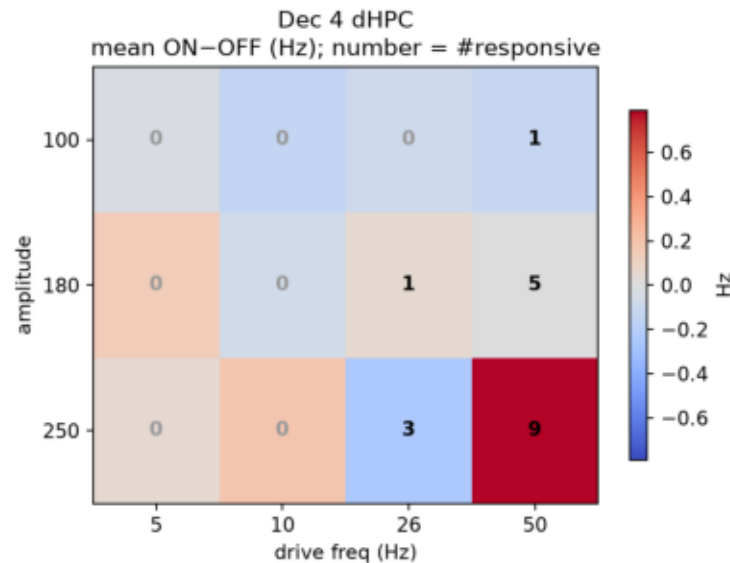
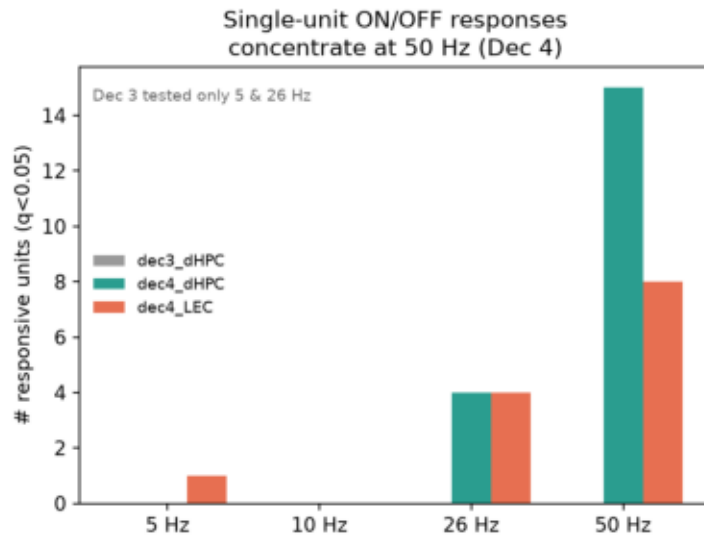
Is the 50 Hz pickup gradient LEC-only? dHPC (left): flat, ~ 0 , no hot channels, dead ch clean. LEC (right): rises toward the dead blocks (128-135, 224-255).



Broken electrodes that CAN'T hear neurons still pick up the 50 Hz — $\sim 6\times$ more than the live ones. So it's pickup.

The real result: individual neurons change their firing at 50 Hz

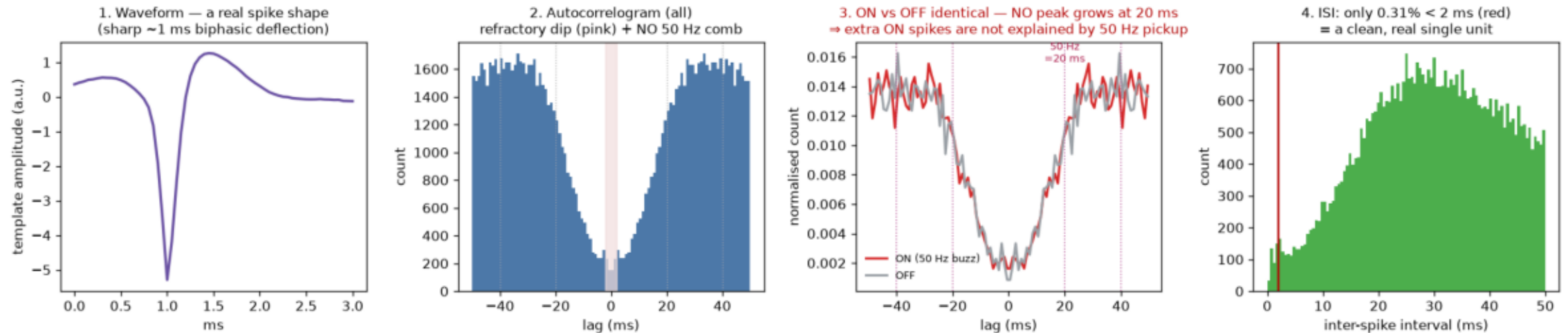
Curated single-unit ON-vs-OFF firing — Dec 3 null at 5/26 Hz; Dec 4 effect at 50 Hz / high amplitude



Single neurons fire differently during the 50 Hz buzz — in BOTH regions. Equipment hum can't fake a neuron firing.

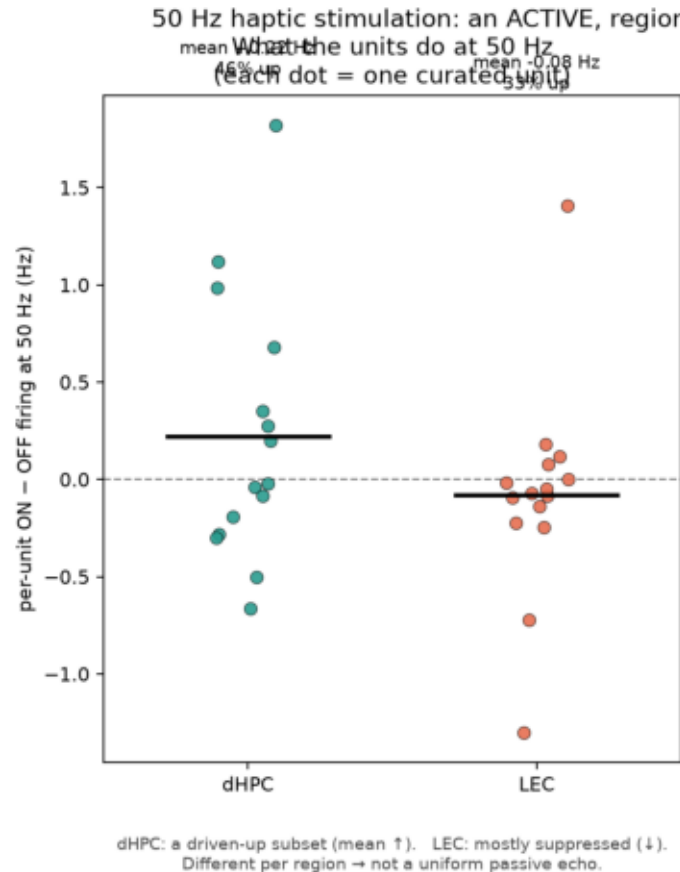
How do we KNOW it's neurons and not the machine? Two filters

UNIT 87 — the 'Phy view', computed: the soft-spot LEC unit is a REAL neuron, and its 50 Hz ON rate increase is not explained by 50 Hz pickup (its autocorrelogram grows no 50 Hz periodicity during ON).
Comprehensive screen: 0/8 up-going units show an ON 50 Hz comb, and 0/8 show an ON ISI<2ms rise $\Rightarrow$ the up-going rate increases are not explained by 50 Hz pickup.



(1) Strip out the slow hum to find the sharp spikes. (2) Check the neuron isn't 'ticking like a clock' at 50 Hz.

Same buzz, opposite reactions — the brain is **PROCESSING**, not echoing



What does the 50 Hz response mean? — the inference ladder

X Is it electrical / mechanical **ARTIFACT**?

NO — sorted single units change their firing RATE; pickup can't do that.

X Is it a **PASSIVE ECHO (brain as a relay)**?

UNLIKELY — regions respond **DIFFERENTLY** (dHPC driven-up subset; LEC suppressed) and the LFP 50 Hz is induced, not phase-locked. A relay would look the same everywhere.

✓ **ACTIVE, region-specific processing?**

SUPPORTED — the 50 Hz input is transformed differently by each circuit (not relayed).

? **Are the regions COORDINATING ("working together")?**

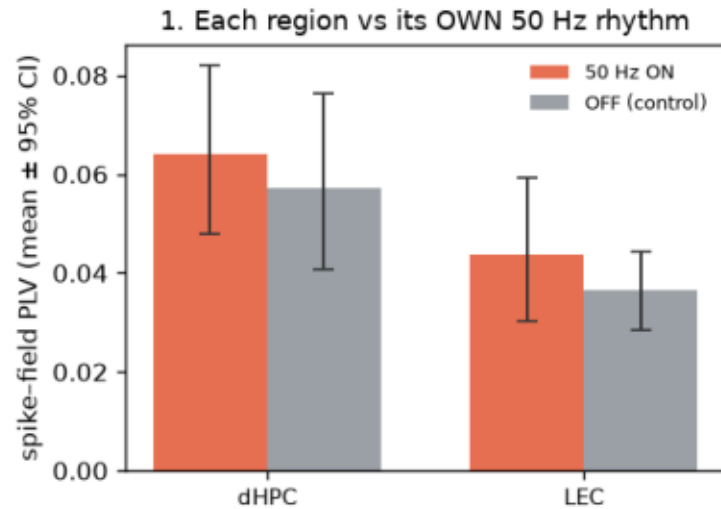
NOT TESTED — needs spike-field coupling + cross-region coherence. We showed each region responds, not that they coordinate.

One region revs a subset of cells **UP**; the other quiets **DOWN**. A passive echo would look the same everywhere.

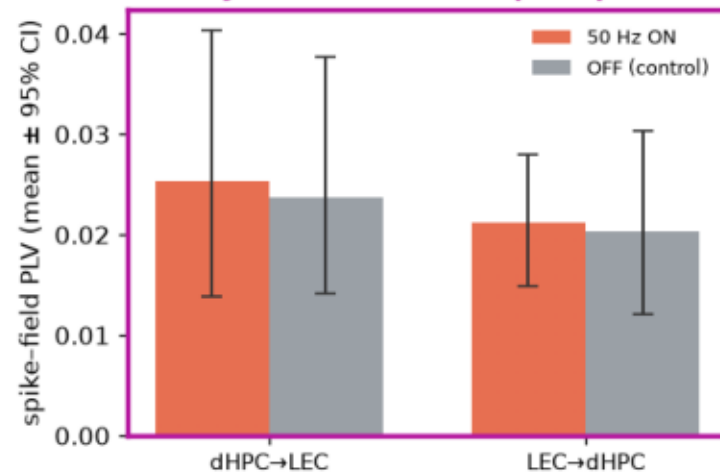
Do the two regions 'work together' at 50 Hz? Not clearly

Do dHPC & LEC 'work together' at 50 Hz? — If they did, panel 2 (cross-region SPIKES) would rise during ON; it does NOT. Panel 3 (LFP) rises but is fooled by a shared signal. ⇒ NO clear coordination.

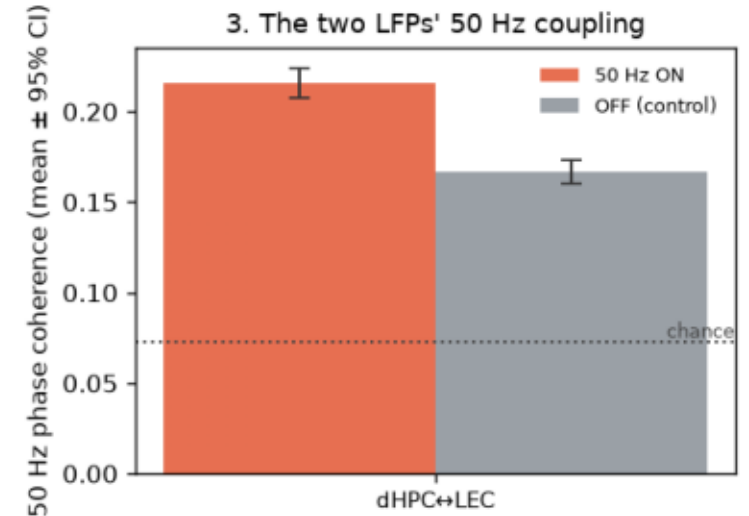
2. SPIKES vs the OTHER region's rhythm (the key test — harder for pickup to fake)



Weak; ON ≈ OFF (CIs overlap).
Baseline gamma coupling,
not stimulus-driven.



Very weak; ON ≈ OFF (CIs overlap).
Neurons do NOT track the other
region → no coordination.



ON > OFF (CIs separate) — looks like
coupling, BUT inflated by a SHARED
signal; spikes (panel 2) don't follow
it → not real coordination.

50 Hz trials, all amplitudes pooled (amp100+180+250, 600 trials). ON = 3 s buzz; OFF = following 3 s gap.

Two orchestras can SOUND synced because of a hallway metronome (a shared hum) — not because they play together.

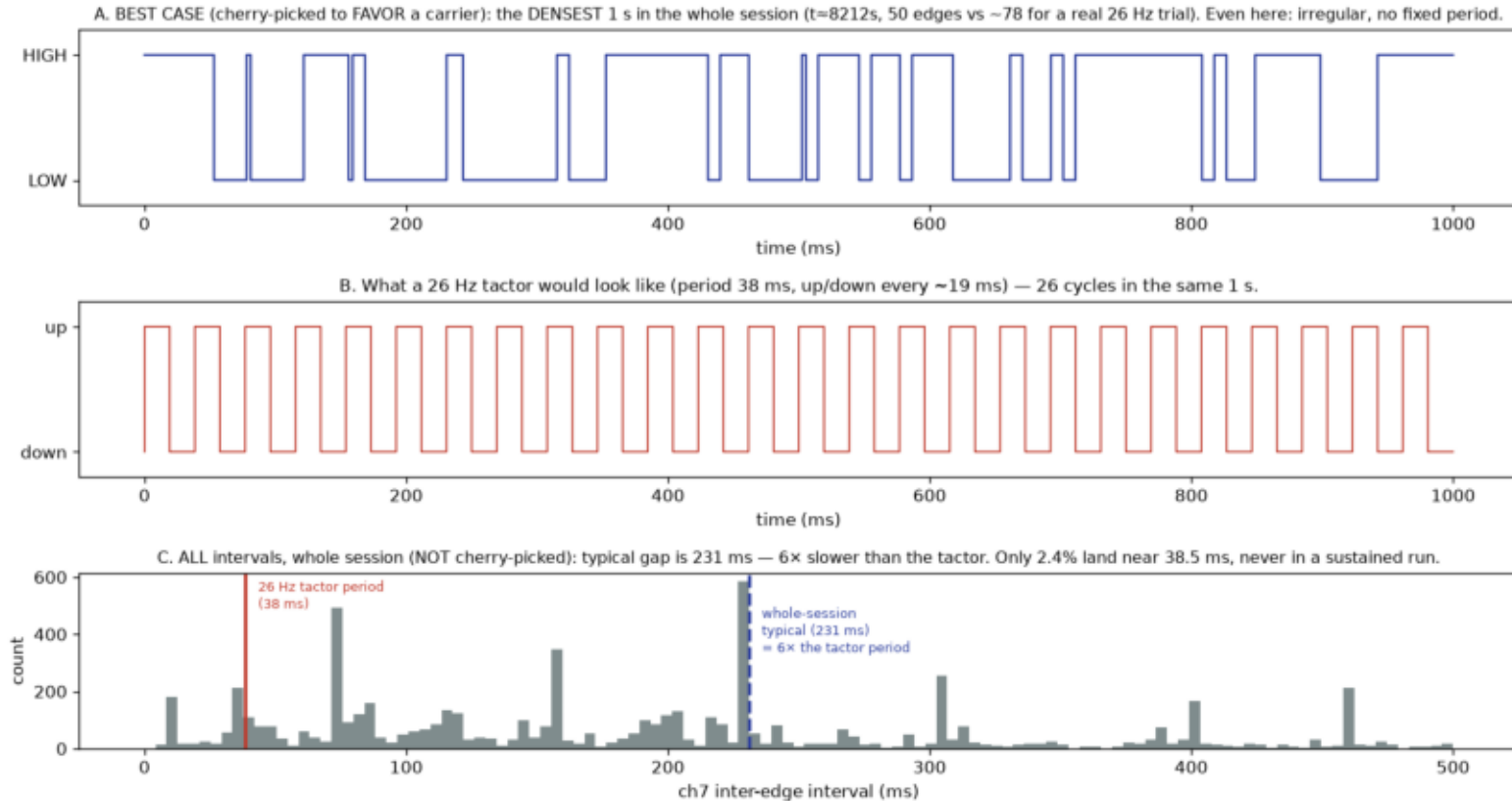
Part 3 — The one thing we couldn't test (and how we fix it)

We could NOT test 'marching in step' — and it's not the brain's fault

It's like judging whether a dancer is on-beat when you forgot to record the music. The reference was missing.

Why: the wire meant to record the buzz never captured it

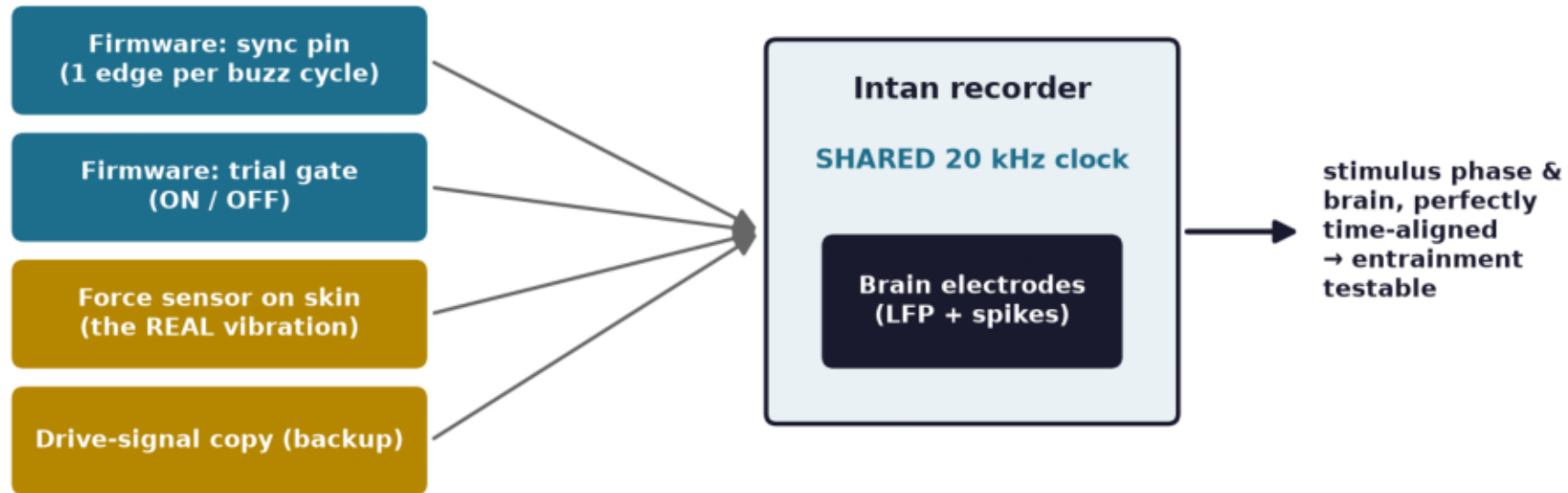
ise. Panel A is the BEST/densest window (cherry-picked to favor a carrier); Panel C is ALL data (typical). ch7 toggles $\sim 4\times/s$; a 26 Hz tactor toggles $26\times/s \rightarrow$ tactor does ~ 6 full



It updated $\sim 4\times/sec$ while the tactor buzzed $26\times/sec$ — like checking a hummingbird's wings 4 times a second.

The fix is concrete — next round records the buzz properly

The fix: record the buzz 3 ways, all on the brain's clock



+ a 2-minute verification recording BEFORE the real session

A per-cycle marker from the firmware + a tiny force sensor that measures the ACTUAL vibration, on the same clock.

The big picture, in plain words

Buzzing the body DOES change the brain — most clearly as single-neuron activity at 50 Hz — but 'driving a brain rhythm' isn't shown yet.

Why it matters

If we can verify entrainment, we move toward wearables that gently and non-invasively steer brain states through touch.

Thank you — questions?

Happy to dig into any figure, the analogies, or the next-round design.